

BingAster at #SMM4H-HeaRD 2025: Identifying Dementia Caregivers on Twitter Using Prompt-Based LLMs and Cognitive Distortion Patterns

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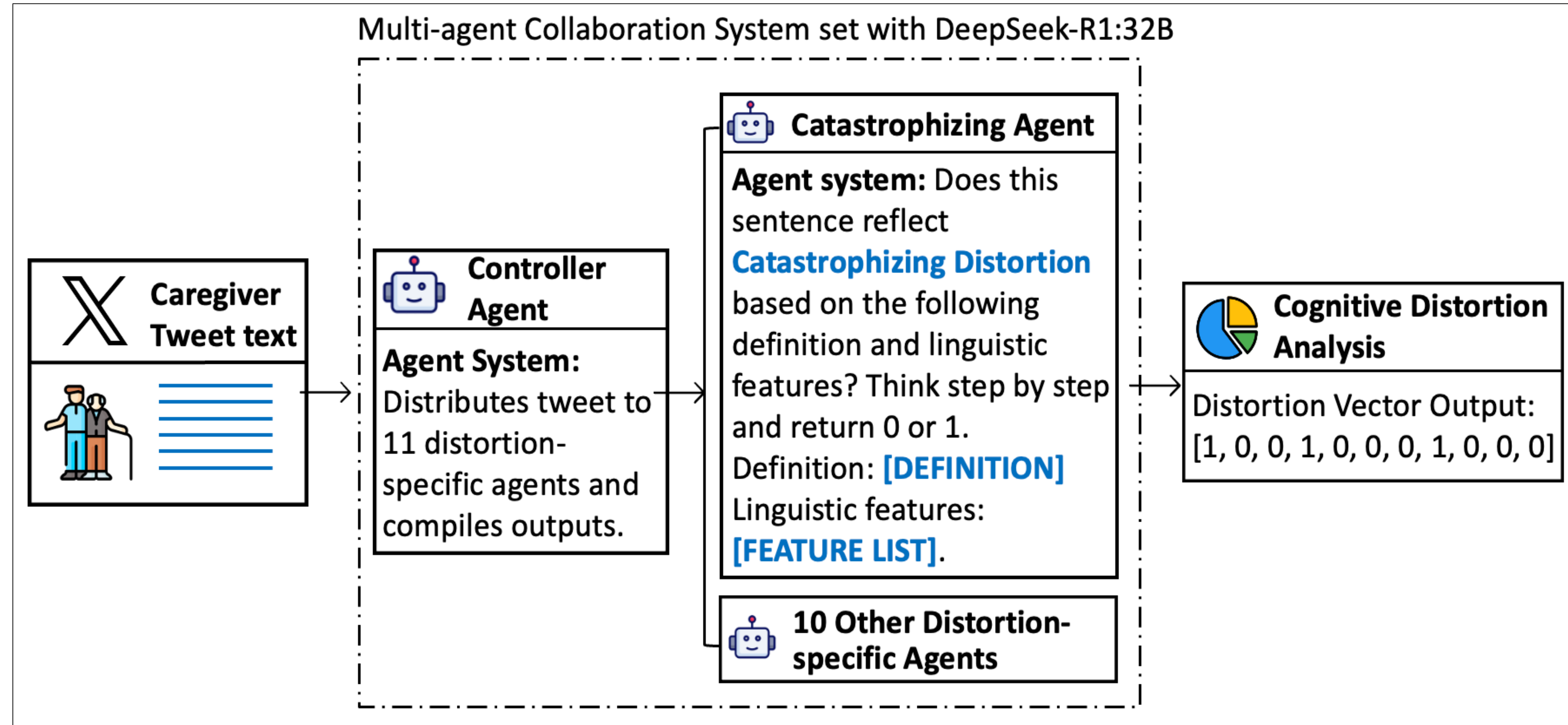
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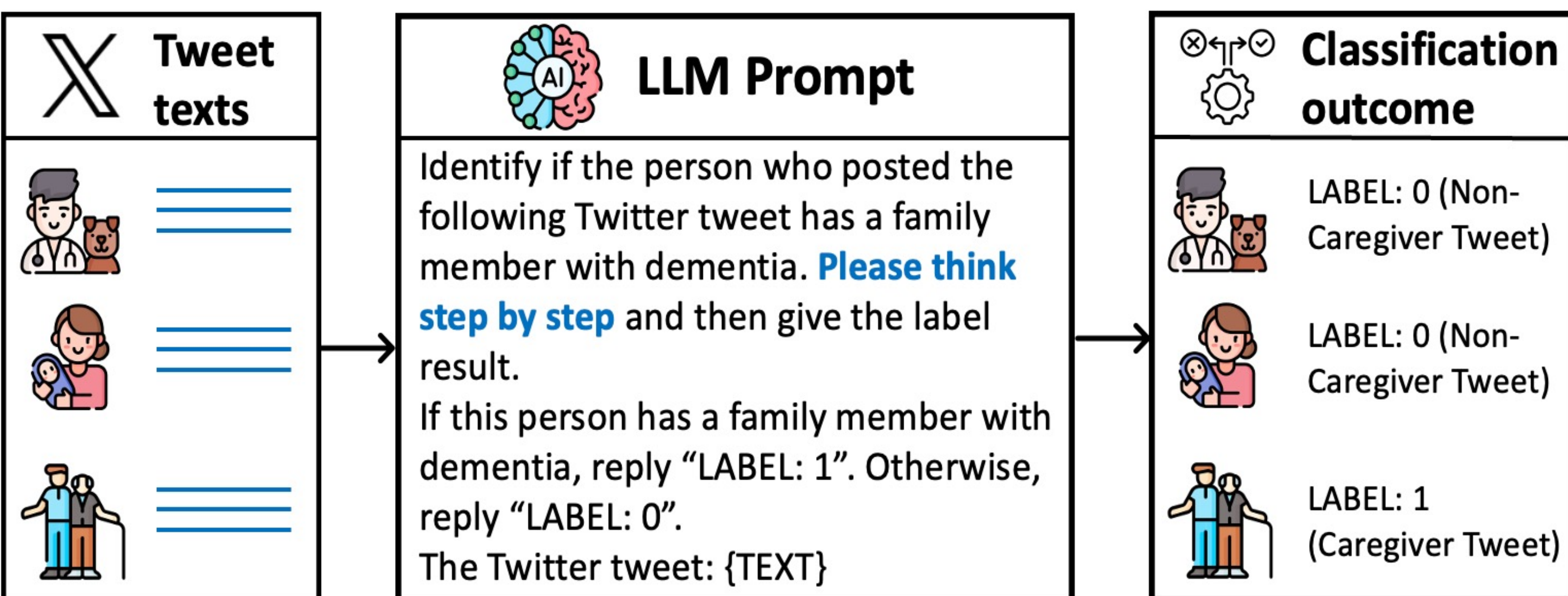
Introduction

- Family members looking after someone with dementia routinely shoulder substantial emotional, physical, and financial strain, which often [reflects in the language they use online](#).
- Our study tackles this identification challenge.** For SMM4H-2025 Task 3, we develop a [prompt-based, zero-shot large-language-model system that flags tweets indicating a family member with dementia](#), laying groundwork for scalable, AI-driven caregiver support.
- We developed a multi-agent system to analyze cognitive distortions** in detected caregiver tweets, [helping further validate our model by uncovering stress-related thinking patterns](#) like catastrophizing and emotional reasoning.

Multi-agent System for Cognitive Distortion Analysis

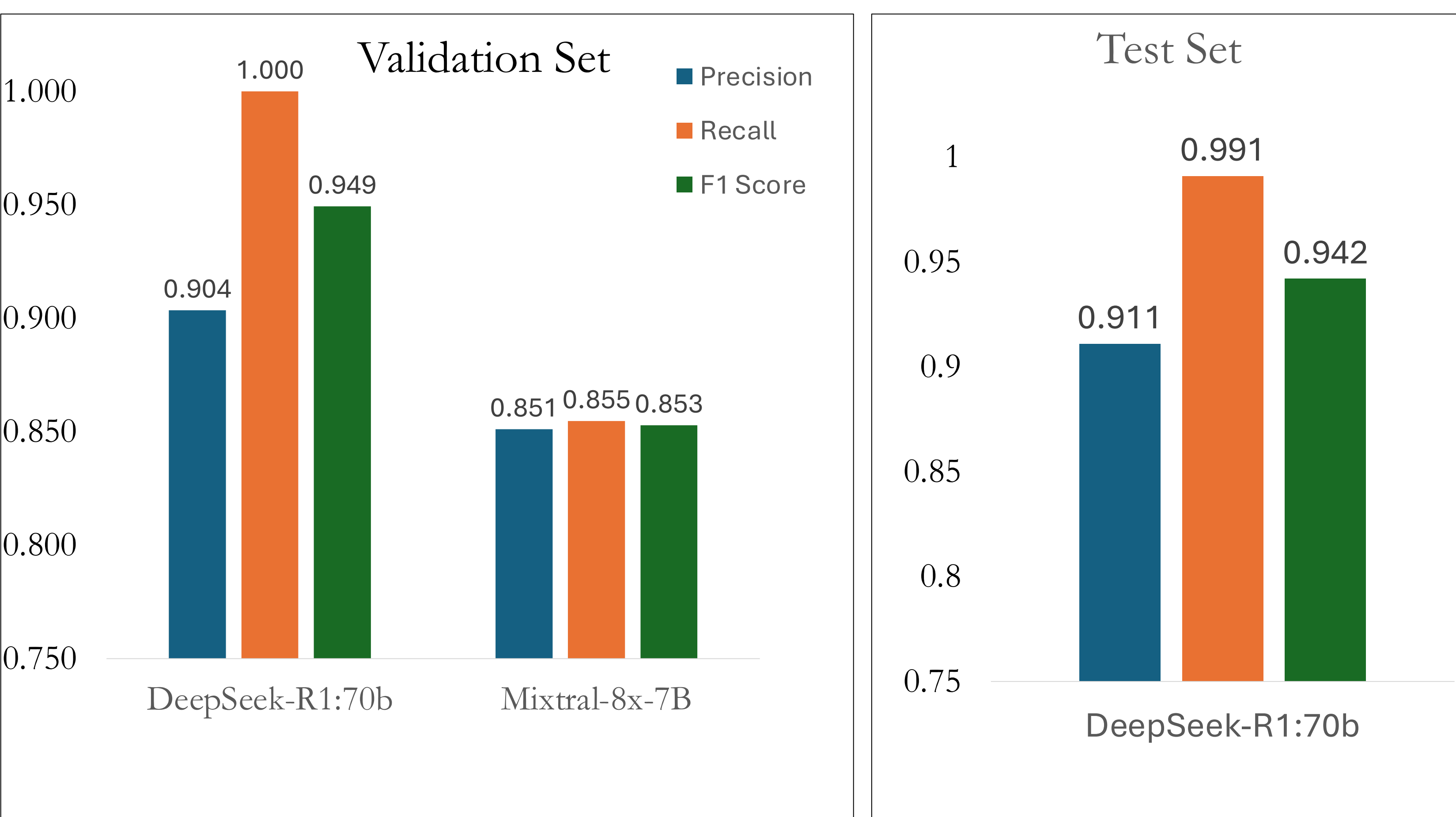


Prompt-based Classification



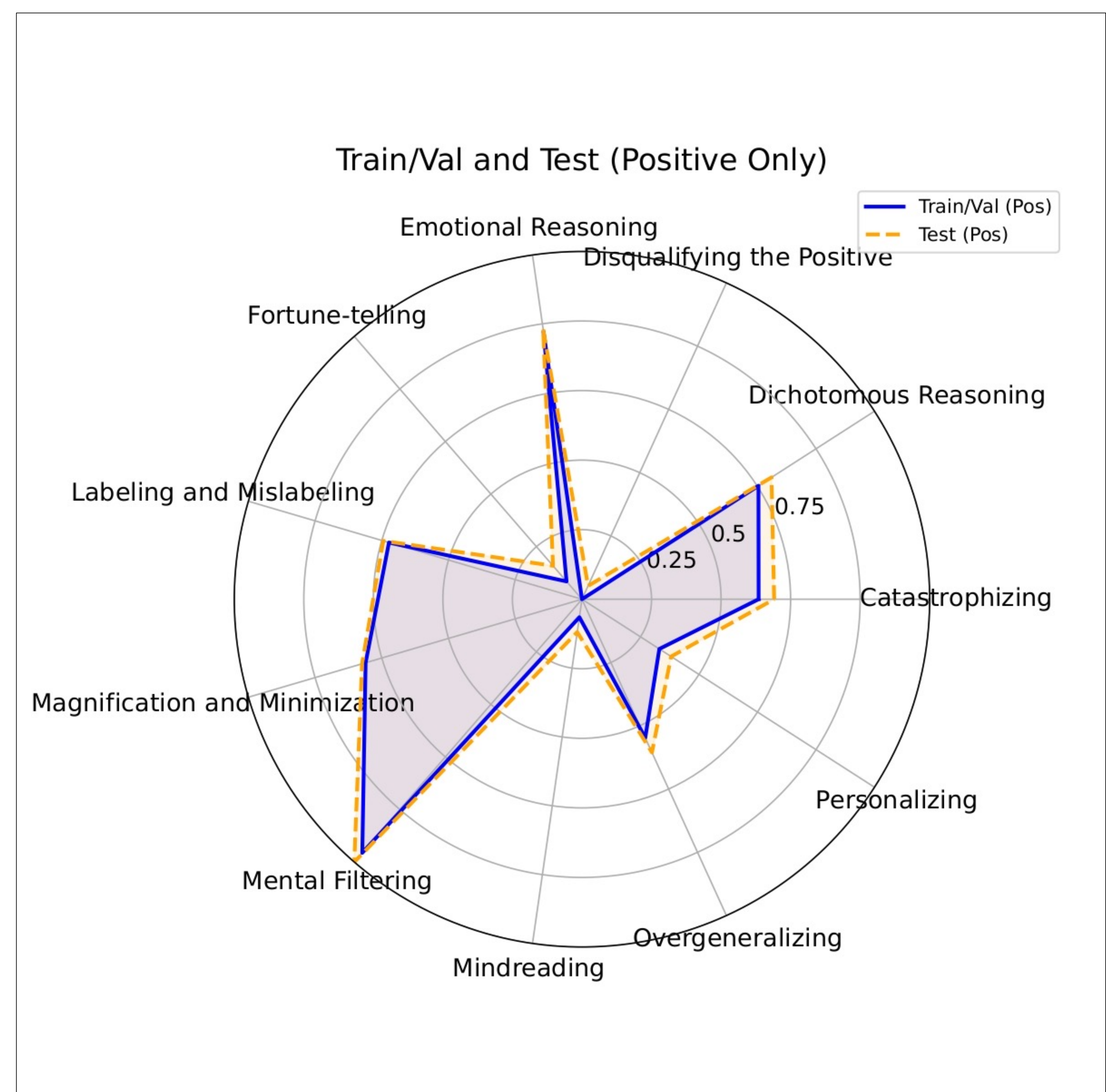
- A zero-shot prompt system is designed to classify tweets as written by a caregiver or not.
- "think step by step" can trigger LLM to conduct [chain-of-thought reasoning](#) in this binary classification task.

Results



- DeepSeek-V1.3 achieved the top F1 score (0.904) on the validation set, outperforming other approaches. It also performed well on the test set (F1 = 0.623), showing strong generalization.

Evaluation by Cognitive Distortion Analysis



- The radar chart shows normalized distortion frequencies for test set (orange) and train/validation set (blue) tweets. Emotional Reasoning, Catastrophizing, and Mental Filtering appear more frequently in caregiver posts.
- These patterns reflect elevated emotional and cognitive stress in caregiver tweets, aligning with known psychological traits. Their consistency across data splits also provides indirect validation that the model is accurately identifying caregivers based on meaningful linguistic signals.

Conclusion and Future Work

- The distortion patterns across training, validation, and test sets suggest that our model effectively generalizes to unseen data while preserving meaningful psychological patterns.
- Future work will focus on improving classification accuracy, expanding cognitive distortion analysis, and designing adaptive online interventions to support caregivers in real time.